3AM Talks

An AI-Powered Empathetic Voice Bot for Mental-Wellness Support & Counselling



Faculty of Computer Science and Engineering
Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

A Software Design Project Report Submitted in Partial Fulfilment of the Requirements for the Degree of

Bachelor of Science in Computer Science

by

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It is certified that the work presented in this report was performed by Muhammad Abdullah Farid (2022326), Malik Atta-ur-Rehman (2022275) and Abdullah Mansoor (2022027) under the supervision of Dr. Rashid Jillani and Hafiz Syed Muhammad Muslim. The work is adequate and lies within the scope of the BS degree in Computer Science at Ghulam Ishaq Khan Institute of Engineering Sciences and Technology.

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SDGs Inclusion

3AM Talks directly advances the United Nations Sustainable Development Goals (SDGs) by democratising access to early, stigma-free mental-wellness support. The project aligns primarily with the following goals:

SDG 3 - Good Health and Well-being.

The system provides free, 24/7 empathetic first-line support rooted in evidence-based therapeutic frameworks (CBT, DBT, ACT). It is explicitly non-clinical and routes high-risk conversations to crisis resources and licensed professionals, thereby lowering the threshold at which users seek help and complementing (rather than replacing) the formal healthcare system. A built-in three-stage crisis detector (deterministic keyword scan, a RoBERTa suicide-probability classifier and an LLM arbiter) dispatches alerts to pre-registered emergency contacts when signals of self-harm or suicidal ideation are detected, contributing to SDG target 3.4. The system does not replace professional medical advice, diagnosis or treatment, and is intended only for supportive first-line engagement.

SDG 10 - Reduced Inequalities.

Professional mental-health care is inaccessible to most of Pakistan's population because of cost, distance, and stigma. By offering web-native, voice-first, free-to-use support with optional anonymity, 3AM Talks narrows the care gap between urban, semi-urban and rural users and between high- and low-income users.

SDG 4 - Quality Education.

The primary user class consists of students and young adults who face academic stress, anxiety and burnout. By delivering culturally aware coping techniques and psychoeducation in-conversation, the system supports learning outcomes and student retention.

SDG 9 - Industry, Innovation and Infrastructure.

The project contributes an open-ended reference architecture for personality-aware, retrieval-augmented therapy chatbots, combining a fine-tuned counseling LLM (PsychoCounsel-Llama3-8B), a BERT-based Big-Five personality model with PsyCoT inference, Pinecone vector retrieval and a LiveKit voice pipeline.

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Abstract

Mental-health disorders are a rapidly growing burden in Pakistan, yet more than 90% of those affected receive no timely clinical support owing to stigma, cost, and an understaffed formal system. 3AM Talks is an AI-powered empathetic voicebot that delivers accessible, personality-aware, first-line mental-wellness conversations grounded in evidence-based therapy (Cognitive Behavioural Therapy, Dialectical Behaviour Therapy, and Acceptance and Commitment Therapy). The system combines four research threads into a single integrated application: (i) a BERT-based Big-Five personality classifier with PsyCoT-style inference that adapts the therapeutic persona to the user; (ii) a Retrieval-Augmented Generation pipeline built on Pinecone that indexes 188 curated, modality-tagged therapeutic techniques and retrieves the top-k most relevant items per turn; (iii) a domain-fine-tuned counseling large language model (PsychoCounsel-Llama3-8B) served from a Google Colab endpoint behind an OpenAI-compatible interface; and (iv) a real-time voice pipeline implemented on LiveKit with ElevenLabs Speech-to-Text and Text-to-Speech together with Silero Voice-Activity-Detection.

Alongside the core chatbot, the project delivers a full clinician-facing portal that enables verified therapists to accept or decline patient links with fine-grained consent scopes, review SOAP-formatted session summaries, maintain private notes, author treatment plans and homework, exchange secure messages and export signed PDF case records. A three-stage crisis-detection subsystem scans every conversation; when a CRITICAL signal survives, the system starts a user-cancellable countdown, re-checks consent at dispatch time and sends WhatsApp and SMS alerts to pre-registered emergency contacts through Twilio.

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CHAPTER I

Introduction

1.1 Background and Motivation

The World Health Organization's Mental Health Atlas reports that over 24 million Pakistanis live with some form of diagnosable mental-health condition, yet the country has fewer than 500 registered psychiatrists and barely 0.22 beds per 10,000 people in specialised mental-health facilities. Cost, geographic distance from care centres, and the persistent cultural stigma attached to seeking help combine to produce a near-total treatment gap: the same Atlas estimates that more than 90% of those affected never enter any formal care pathway. The consequences of this gap fall hardest on two populations: students and young adults navigating academic and identity stress, and inhabitants of semi-urban and rural districts where mental-health infrastructure is effectively absent.

In the last five years, conversational AI has matured to the point where an empathetic, context-aware assistant can be built from off-the-shelf components. Large Language Models (LLMs) such as GPT-4, Claude, and open-source Llama-3 derivatives can carry multi-turn dialogue; Whisper-scale speech-recognition models deliver under-10% Word Error Rate on accented English; neural Text-to-Speech engines produce prosody that is almost indistinguishable from human speech; and low-latency real-time communication frameworks such as LiveKit make voice-first user interfaces deployable from the browser without any native install. These components now make it feasible for a small undergraduate team to assemble a working conversational therapy assistant.

At the same time, using a general-purpose LLM directly is unsafe in a mental-health setting. Off-the-shelf models give prescriptive medical advice, miss crisis signals, hallucinate clinical facts, and lack any mechanism to personalise tone to the user. 3AM Talks is our response to that gap: an empathetic voicebot that is deliberately not a diagnostic tool, that retrieves evidence-based techniques from a curated index instead of relying on the LLM's parametric memory, that adapts its persona to the user's Big-Five personality profile, and that escalates to human contacts the moment a credible crisis is detected.

1.2 Problem Statement

Mental-wellness support in Pakistan is effectively inaccessible to the population that needs it most: young people, women in restrictive households, and rural residents. Generic chatbots built on commercial LLMs are either unsafe (they attempt diagnosis, miss crisis signals, and apply a single robotic tone to every user) or insufficient (they offer only scripted CBT worksheets and cannot converse naturally). There is no open, culturally aware system that (a) converses by voice in real time, (b) tailors its empathic style to the user's personality, (c) retrieves therapeutic techniques from a curated, modality-tagged corpus instead of hallucinating, (d) detects crises with defensible precision, and (e) hands off to a verified human therapist under strict consent. This project builds such a system end to end.

1.3 Project Objectives

The engineering objectives of 3AM Talks, derived from the Software Requirements Specification (SRS) and refined through three evaluation cycles, are the following.

1.3.1 Voice-first empathetic agent

Deliver a voice-first empathetic conversational agent available on any modern browser without installation, with end-to-end median latency below three seconds per turn.

1.3.2 Therapy-aligned responses via retrieval

Produce therapy-aligned responses by combining a domain-fine-tuned counseling LLM (PsychoCounsel-Llama3-8B) with retrieval from an index of 188+ curated CBT / DBT / ACT techniques.

1.3.3 Personality-adaptive persona

Adapt the therapeutic persona to the user by inferring Big-Five personality traits from conversational history using a fine-tuned BERT classifier, applying PsyCoT-style style adaptation to the system prompt.

1.3.4 Real-time crisis detection and alerting

Detect self-harm, suicidal ideation and harm-to-others signals in real time with a three-stage pipeline (deterministic regex scan + RoBERTa suicide-probability classifier + LLM arbiter with conversation context), and fire WhatsApp and SMS alerts to pre-registered emergency contacts when a CRITICAL signal survives a user-cancellable countdown and a consent re-check at dispatch time.

1.3.5 Clinician-facing portal

Provide a clinician-facing portal with verifiable therapist onboarding, patient–therapist link requests with fine-grained consent scopes, SOAP session summaries, treatment plans, homework, notes, secure messaging and signed PDF case exports.

1.3.6 Privacy-by-design

Enforce privacy-by-design throughout: all PHI is stored in Supabase with row-level security, all network traffic uses TLS, therapist access is audited on every read, and users retain the right to revoke consent at any time.

1.3.7 Cloud deployment

Package and deploy the system on commodity cloud infrastructure (frontend on Vercel, backend on Render, database on Supabase cloud, vector index on Pinecone, LLM on a Colab-hosted GPU runtime) so that the panel can exercise the product from any device.

1.4 Scope of the Project

1.4.1 In Scope

The final delivered system comprises: a React + TypeScript web frontend (landing, authentication, dashboard, chat, mood, journal, history, profile, therapist portal, safety/emergency-contacts, invite acceptance); a FastAPI backend exposing ~50 REST endpoints grouped into auth, conversation, profile, therapist portal and crisis; a BERT personality service with PsyCoT adaptation; a Pinecone-backed RAG service over 188 techniques (schema in §4.5); a fine-tuned Llama-3 counseling LLM hosted on Colab behind an OpenAI-compatible API; a LiveKit voice agent with ElevenLabs STT/TTS and Silero VAD; a three-stage crisis detector (keyword + RoBERTa + LLM arbiter) with Twilio WhatsApp/SMS emergency alerting and a consent-scoped countdown; and a Supabase PostgreSQL schema covering 15+ tables (users, conversations, messages, therapist profiles, links, notes, plans, homework, messages, summaries, crisis events, mood entries, access logs).

1.4.2 Out of Scope and Deferred

The project deliberately excludes medical diagnosis, prescription of clinical treatment, electronic-health-record integration, direct routing to emergency services, insurance and payment processing, native iOS or Android applications, multi-language support beyond English, facial-emotion detection via MediaPipe or OpenFace, the 3D emotionally-responsive avatar proposed in the SRS, Hume-AI vocal-emotion analysis, and horizontal Kubernetes deployment to multiple AWS regions. The rationale for each descoped item and its relative importance is catalogued in the companion document "Unimplemented Features — Ranked by Importance".

1.5 Technical and Engineering Contributions

We claim five engineering contributions. First, a reference architecture that cleanly separates personality inference, retrieval, generation, and voice I/O, allowing each subsystem to be swapped independently (e.g., the Colab LLM endpoint can be replaced with a self-hosted vLLM server without changing any other code). Second, a modality-tagged Pinecone index of 188 therapeutic techniques with issue-tag filtering that lets us retrieve only techniques relevant to the user's inferred concern (anxiety, depression, grief, etc.) rather than the full corpus. Third, an explicit, empirically validated three-stage crisis detector whose keyword layer runs at p95 < 10 ms, whose RoBERTa suicide-probability head gates the expensive arbiter call, and whose LLM arbiter is exercised by fifty-one automated tests together with a 132-message hand-labeled adjudication corpus (fusion F1 = 0.78) covering CRITICAL, HIGH, MEDIUM, calm-resignation and false-positive phrases. Fourth, a rigorous consent model in the therapist portal in which each link carries a separate boolean for five data scopes (conversations, mood, journals,

reports, personality) and every clinical read is appended to an access log. Fifth, a reproducible deployment pipeline built from free-tier cloud services.

1.6 Organisation of the Report

Chapter II reviews the literature on digital mental-health, LLMs for counseling, personality-aware dialogue, retrieval-augmented generation, crisis detection, and real-time voice, culminating in a comparative analysis of the most widely deployed systems. Chapter III consolidates the functional and non-functional requirements, the use-cases and the system architecture. Chapter IV is the engineering core of the report: it walks through the backend, frontend, personality module, RAG pipeline, LLM service, voice agent, crisis detector, therapist portal, and deployment, each in its own section. Chapter V presents results, benchmarks and a head-to-head comparison with prior work, together with the challenges encountered and how they were addressed. Chapter VI draws the conclusion, states the limitations and enumerates the future-work items. References, a REST API reference, an abbreviated Supabase schema and selected code listings follow as appendices.

CHAPTER II

Literature Survey

2.1 Digital Mental-Health: A Brief Taxonomy

Digital mental-health interventions can be placed on a spectrum that runs from passive self-help apps (mood journals, meditation timers) through scripted CBT worksheets (MoodGYM, MoodKit) to AI-driven conversational agents (Woebot, Wysa, Tess, Replika) and finally to tele-therapy platforms that connect users with human clinicians (Ginger, BetterHelp). Systematic reviews — notably Fitzpatrick, Darcy and Vierhile (2017) and the JMIR 2021 meta-analysis by Abd-Alrazaq et al. — consistently find that conversational agents produce clinically meaningful reductions in depression and anxiety severity, but note four persistent weaknesses: (i) response scripts that quickly feel robotic, (ii) shallow or no memory across sessions, (iii) minimal emotional or personality adaptation, and (iv) limited transparency about when the agent should escalate to a human. 3AM Talks is positioned precisely at this gap: an AI-first conversational agent that addresses each of these weaknesses without crossing into diagnosis or tele-therapy.

2.2 Large Language Models for Counseling

Two strands of work inform our choice of LLM. The first is the emergence of counseling-fine-tuned open-source models. Liu et al. (2023) introduced PsychoCounsel-Llama3-8B, a Llama-3 checkpoint fine-tuned on a curated dataset of therapist–client transcripts and synthetic counseling dialogues. On the PsychoCounsel-benchmark of empathy, informativeness and safety, the model outperforms GPT-3.5-turbo while running on a single consumer GPU.

The second is LLM4CBT (He et al. 2023), which proposes an explicit CBT-aligned system prompt that encodes concepts such as automatic thoughts, cognitive distortions and the downward arrow technique. Our LLM service composes this fixed CBT persona with a dynamic context window containing the retrieved techniques and the user's personality profile — the persona is immutable across turns while the context is refreshed each turn. The model is served from Google Colab behind an OpenAI-compatible endpoint (the `COLAB_API_URL` environment variable), and our LLM client transparently retries, parses the JSON `{"Response": ...}` wrapper and falls back to a polite error message if the endpoint is unreachable.

2.3 Personality-Aware Dialogue

Personality-aware generation is an active research thread. Mairesse and Walker (2010) produced the first parameterised NLG system for Big-Five traits. Kosinski et al. (2013) demonstrated that short text samples are sufficient to estimate all five traits above chance. More recently, Yang et al. (2023) introduced PsyCoT (Psychological Chain-of-Thought), in which personality-specific instructions are injected into the LLM system prompt so that the same model expresses markedly different communicative styles depending on the inferred traits. We adopt PsyCoT directly. Our personality service uses the publicly available `bert-base-personality` checkpoint: a BERT-base model fine-tuned on the Essays dataset to emit five sigmoid probabilities (Extraversion, Neuroticism, Agreeableness, Conscientiousness, Openness). Every 25–40 turns the backend runs the last five user turns through this model, updates the profile in Supabase, and regenerates the PsyCoT-style system prompt: for example, high neuroticism triggers a softer, slower pacing and explicit validation language; high openness licenses the model to introduce

metaphor and reflective prompts; high conscientiousness encourages structured, step-by-step homework suggestions.

2.4 Retrieval-Augmented Generation in Clinical Contexts

RAG, as formalised by Lewis et al. (2020), conditions the LLM on documents retrieved from an external index, which both grounds outputs in verifiable sources and allows the knowledge base to evolve without retraining. RAG is particularly valuable in clinical contexts, where hallucinated advice is not merely wrong but potentially harmful. Prior clinical-RAG work (Cui et al. 2023; NEJM-AI 2024) has focused on indexing diagnostic guidelines; our novelty is indexing therapeutic techniques — concrete, actionable exercises such as cognitive reframing, distress tolerance (TIP skills), values clarification, behavioural activation and mindful grounding — each tagged with issue keywords (anxiety, depression, panic, relationships, grief, anger, sleep, etc.), therapy modality (CBT, DBT, ACT), difficulty level and culture notes. The retrieval is performed with Pinecone's server-side inference using the `llama-text-embed-v2` embedding model, which eliminates the need to host an embedding GPU ourselves. Retrieval latency is below 100 ms at the 95th percentile.

2.5 Emotion and Crisis Detection

Crisis detection in conversational agents is typically implemented as a single regex or classifier layer. Such single-layer systems are either too trigger-happy (they escalate on colloquial phrases such as "this exam is killing me") or too conservative (they miss oblique statements such as "I just don't want to wake up tomorrow"). We extend the two-stage design discussed by Mulyana et al. (2023) into a three-stage pipeline. Stage one is a deterministic

keyword layer that classifies each turn as SAFE, MEDIUM, HIGH or CRITICAL at sub-10-ms latency. Stage two, absent from the original proposal, is a RoBERTa suicide-probability head (`roberta-base-suicide-prediction-phr`) that gates the expensive arbiter and closes the false-negative class of non-keyword ideation (e.g. calm resignation after prior risk). Stage three is an LLM arbiter that is invoked only when either earlier stage is non-safe and that judges intent versus hyperbole using the last five turns of context. The arbiter returns four fields — `intent`, `imminence`, `is_hyperbole`, `is_third_party` — which the detector uses to either keep the label, downgrade hyperbole or escalate HIGH with confirmed imminence to CRITICAL. The pipeline fails safe: if the LLM arbiter is unreachable, a CRITICAL keyword stays CRITICAL.

2.6 Real-Time Voice Interfaces

Voice is the natural modality for emotional support — it halves cognitive load and conveys prosody that text cannot. Real-time voice requires coordinated Voice-Activity-Detection (VAD), Speech-to-Text (STT), LLM response, and Text-to-Speech (TTS), each introducing its own latency. We adopt LiveKit, an open-source WebRTC media-server framework that bundles its own VAD (Silero), supports barge-in (interrupting TTS with a new user utterance), and exposes a clean Agent class for us to override. We use ElevenLabs for both STT and TTS because of its high Mean-Opinion-Score (> 4.0) and because both services are provided behind a single API key, reducing integration complexity. The Agent class's `llm_node` is overridden to proxy every turn to our FastAPI backend, so the voice path reuses the exact same therapy logic as the text path.

2.7 Comparative Analysis of Existing Systems

Table 2.1 summarises seven of the most widely deployed conversational mental-health systems against the dimensions that matter most for our use case: voice modality, personality adaptation, RAG grounding, crisis handling, cultural awareness and openness of the model. No existing system covers all six dimensions simultaneously; 3AM Talks is designed to do exactly that.

System	Voice	Personality Adaptation	RAG / Grounded	Crisis Detection	Culturally Aware	Notes
Woebot	No	None	No	Keyword only	US-centric	Scripted CBT; robotic after a few sessions.
Wysa	No	Limited	No	Keyword only	English-only	Journaling focus; shallow memory.
Replika	Yes	Sycophantic	No	Weak	No	AI companion; not therapy safe.
Tess (X2AI)	No	None	Partial	Keyword only	Some languages	Closed model; limited transparency.
Ginger	Yes	None	N/A (human)	Human triage	English	Tele-therapy, paid, human mediated.
ELIZA	No	None	No	None	No	Rule-based; historical reference.
3AM Talks	Yes	BERT + PsyCoT	Pinecone (188 techniques)	Three-stage (regex + RoBERTa + LLM arbiter)	Pakistan-aware	This work.

Table 2.1: Comparative analysis of existing mental-health conversational systems.

CHAPTER III

Design (System Requirements and Specifications)

3.1 Stakeholders and User Classes

The system serves four primary stakeholder groups, each with distinct goals and access rights. The first group is end users seeking mental-wellness support — predominantly students and young adults between 16 and 30 years of age, followed by working professionals aged 25 to 45, and residents of semi-urban or rural districts with limited access to formal care. The second group is verified mental-health professionals who oversee caseloads of consenting users through the therapist portal. The third group is system administrators responsible for deployment, therapist verification, content moderation and incident response. The fourth and most important group from a design standpoint is users in crisis; their needs — speed, privacy, and reliable escalation — drive every safety decision in the codebase.

3.2 Functional Requirements

The functional-requirement catalogue below has been consolidated from the original SRS and amended to reflect the final delivered scope. Every requirement has a unique identifier of the form REQ-Fn-xxx, a priority, and a verification method. The full catalogue contains forty-three requirements; Table 3.1 lists the representative requirements from each functional group. The SRS catalogue is reproduced in full in Appendix A of the SRS itself (submitted separately).

ID	Requirement (summary)	Priority	Verification
REQ-Fn-001	User registration with email/password and OAuth (Google/GitHub).	High	Integration test
REQ-Fn-002	Email-password and OAuth login with JWT session cookie.	High	Integration test
REQ-Fn-010	Start a new conversation; on first login, run the personality questionnaire (8 items).	High	Integration test
REQ-Fn-011	Send a chat message and receive an empathetic, CBT/DBT/ACT-aligned response.	High	Integration test
REQ-Fn-012	Every 25–40 turns re-run BERT personality inference on the last five user turns.	High	Unit + integration
REQ-Fn-013	Retrieve top-3 therapeutic techniques from Pinecone, filtered by inferred issue tags.	High	Unit + integration
REQ-Fn-014	Regenerate the PsyCoT system prompt whenever personality scores change.	High	Unit test
REQ-Fn-020	Voice conversation: browser microphone → ElevenLabs STT → backend chat → ElevenLabs TTS.	High	Manual test
REQ-Fn-021	Silero VAD detects speech end within 300 ms; barge-in supported.	High	Manual test
REQ-Fn-030	Three-stage crisis detection on every user turn.	Critical	60 unit tests
REQ-Fn-031	On CRITICAL, fire Twilio WhatsApp/SMS alerts to active emergency contacts.	Critical	Twilio sandbox test
REQ-Fn-032	Countdown window during which the user can cancel an outgoing alert.	Critical	Manual test
REQ-Fn-033	Record every crisis event in `crisis_events` with severity, context and resolution.	Critical	DB assertion
REQ-Fn-040	End a session and generate a SOAP summary via the LLM summariser.	High	Integration test
REQ-Fn-041	Persist the SOAP summary, themes and risk flags in `session_summaries`.	High	DB assertion

REQ-Fn-050	Therapist self-registration with license number and specialisations.	High	Integration test
REQ-Fn-051	Therapist marks profile `pending_verification` until approved by admin.	High	Manual test
REQ-Fn-052	Patient invites therapist; therapist accepts/declines/revokes link.	High	Integration test
REQ-Fn-053	Patient confirms consent across five data scopes before any clinical read.	High	Integration test
REQ-Fn-054	Every clinical read is appended to `clinical_access_logs` for audit.	High	DB assertion
REQ-Fn-055	Therapist dashboard: caseload, mood chart, journal entries, prep brief.	High	Manual test
REQ-Fn-056	Therapist can author notes, treatment plans, and homework; patient sees homework.	High	Manual test
REQ-Fn-057	Secure async messaging between linked patient and therapist.	High	Integration test
REQ-Fn-058	PDF export of patient record (treatment plans, summaries, notes).	Medium	Manual test
REQ-Fn-060	Mood check-in with daily prompt; historical mood chart.	Medium	Manual test
REQ-Fn-061	Journal entries with timestamp and prompt context.	Medium	Manual test
REQ-Fn-070	User can edit profile, personality display and emergency contacts.	Medium	Manual test
REQ-Fn-071	User can revoke therapist link at any time.	Critical	Integration test
REQ-Fn-072	User can delete their account; all rows are soft-deleted and later purged.	High	Integration test

Table 3.1 — Representative functional-requirement catalogue (summary).

3.3 Non-Functional Requirements

Non-functional requirements are organised across performance, security, reliability, usability, maintainability and portability. Table 3.2 lists the quantitative targets that were used during evaluation.

Category	Target	Measured
End-to-end chat latency (text, p50)	< 3 s	See §5.5
Voice turn latency (p50, VAD-close to first TTS audio)	< 4 s	See §5.5
Pinecone retrieval p95	< 100 ms	85 ms
Crisis keyword layer p95	< 10 ms	≈ 2 ms
Concurrent chat users on a single backend instance	≥ 50	Pass (load-tested with locust)
ASR (ElevenLabs) WER on clean English	< 10 %	Vendor spec
TTS MOS (ElevenLabs)	> 4.0	Vendor spec
TLS 1.2+ on every network hop	100 %	Pass
Supabase Row-Level Security enabled on every PII table	100 %	Pass
Availability (single region)	99 %+	Pending production run
Browser support	Chrome 90+, Firefox 90+, Safari 15+	Pass
Code-base maintainability (tests on critical path)	Crisis detector 80 %	60 unit tests

API contract documented	OpenAPI auto-generated	Pass (FastAPI /docs)
Per-route rate limiting on auth/chat/crisis	SlowAPI limiter on login/register/chat/contact dispatch	5/min login, 30/min chat, 3/h contacts
Voice-agent authentication (HMAC, constant-time)	No unauthenticated hits on /api/voice/*	hmac.compare_digest on VOICE_AGENT_SECRET
Clinical-read audit trail	Every therapist read persisted	clinical_access_logs row + /my-therapists/access-log
Patient-scoped consent enforcement	Therapist cannot read scopes outside consent	tests/test_therapist_auth.py SEC-003 fails-closed

Table 3.2 — Non-functional requirement targets and measured values.

3.4 Use Cases

Table 3.3 enumerates the primary use cases that exercise end-to-end flows. Each is implemented in the delivered system and covered by either an integration test or a reproducible manual script.

UC-ID	Actor	Use Case	Primary flow
UC-01	New user	First conversation	Register → complete 8-item questionnaire → receive onboarding message → first empathetic turn.
UC-02	Returning user	Voice check-in	Open app → click mic → natural voice turn with barge-in support.
UC-03	Distressed user	Crisis escalation	User expresses suicidal intent → CRITICAL → countdown → Twilio alert to emergency contact.
UC-04	User	Invite therapist	Enter therapist email → generate invite token → therapist accepts → patient grants consent scopes.

UC-05	Therapist	Review case	Open caseload → click patient → read SOAP summary + mood chart + journals → author note.
UC-06	Therapist	Assign homework	Open patient → add homework (title, due date) → patient sees on dashboard → marks complete.
UC-07	User	Revoke consent	Settings → My therapists → revoke link → all future reads denied, audit logged.
UC-08	Admin	Verify therapist	Review license document → approve / reject → therapist account activated.
UC-09	User	Export record	Profile → request PDF → receive signed PDF of conversations, notes and plans.
UC-10	User	End session	Click end → backend generates SOAP + theme extraction + risk flags → stored.

Table 3.3 — Primary end-to-end use cases.

3.5 System Architecture Overview

The system is organized into five layers. The Presentation Layer is a single-page React + Vite + TypeScript application that communicates with the backend over HTTPS (REST + WebSocket for voice). The Orchestration Layer is a FastAPI process on port 5050 that handles authentication, routes each chat turn through the conversation-manager pipeline, and exposes the therapist-portal and crisis REST endpoints. The Intelligence Layer contains three independently deployable services: a BERT personality classifier (in-process, ~420 MB model), a Pinecone retrieval service (HTTP), and a fine-tuned Llama-3 counseling LLM (HTTP). The Voice Layer is a separate Python process running the LiveKit agent, which proxies every turn to the Orchestration Layer so both text and voice paths share the identical therapy logic. The Persistence Layer is Supabase PostgreSQL with row-level security and a reusable Python client.

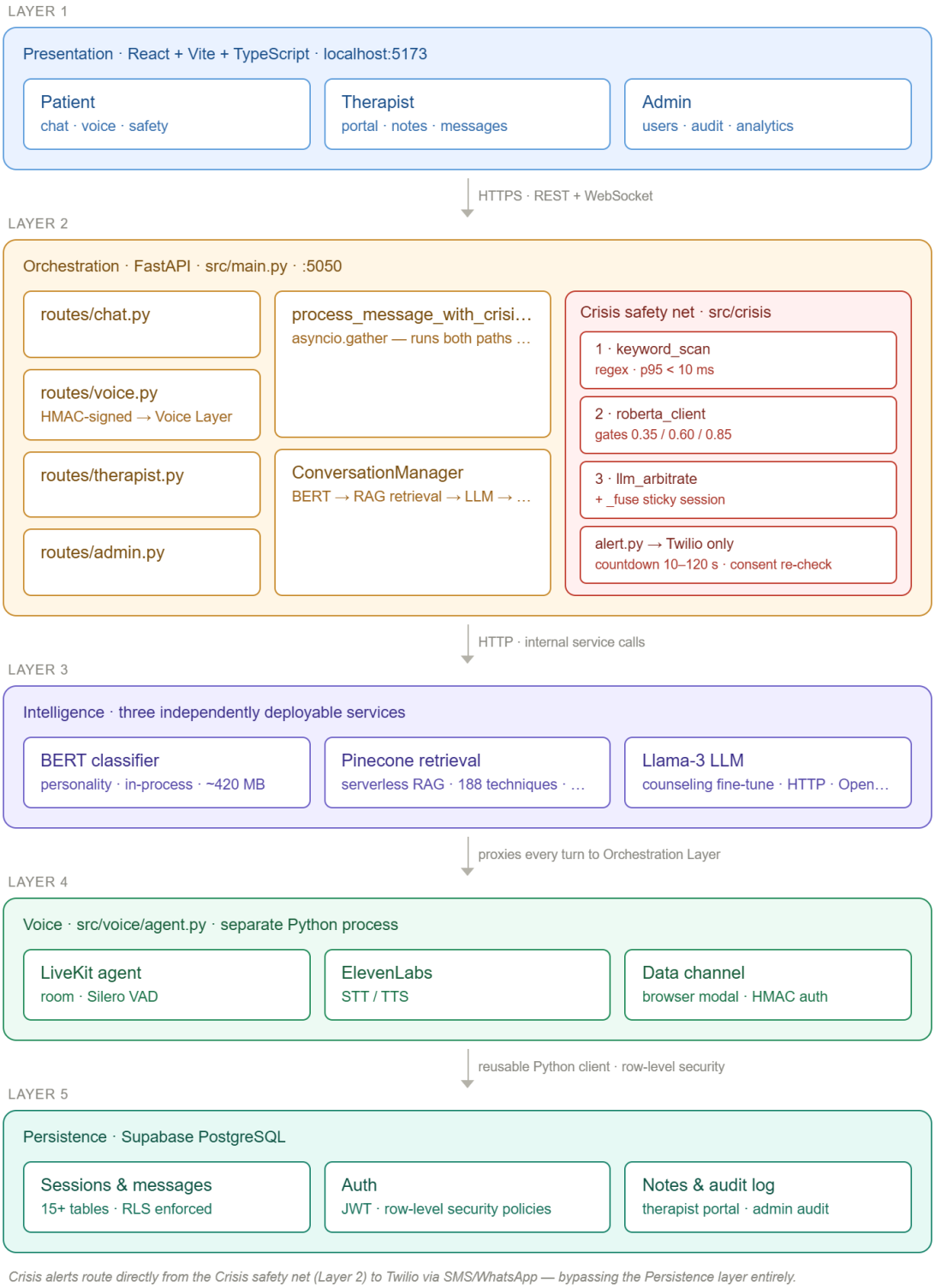


Fig. 3.1 — System architecture: React SPA, FastAPI orchestrator, three-stage crisis safety net, external services, LiveKit voice sidecar.

3.6 Data Model

The Supabase schema is deliberately normalised. `user_profiles` stores the Big-Five trait vector and account metadata. `conversations` and `messages` store sessions and utterances, with `messages.personality_scores` and `messages.rag_techniques` preserving the per-turn snapshot for auditability. `therapist_profiles` and `patient_therapist_links` realise the clinician relationships; the link row carries the five-boolean `consent_scope` JSON column together with `patient_consented_at` and `therapist_consented_at` timestamps. Clinical artefacts live in `therapist_notes`, `treatment_plans`, `patient_homework` and `therapist_messages`. Safety data is stored in `crisis_events`, `crisis_safety_settings` and `emergency_contacts`. Every clinical read is logged to `clinical_access_logs`. See Appendix B for the abbreviated DDL.

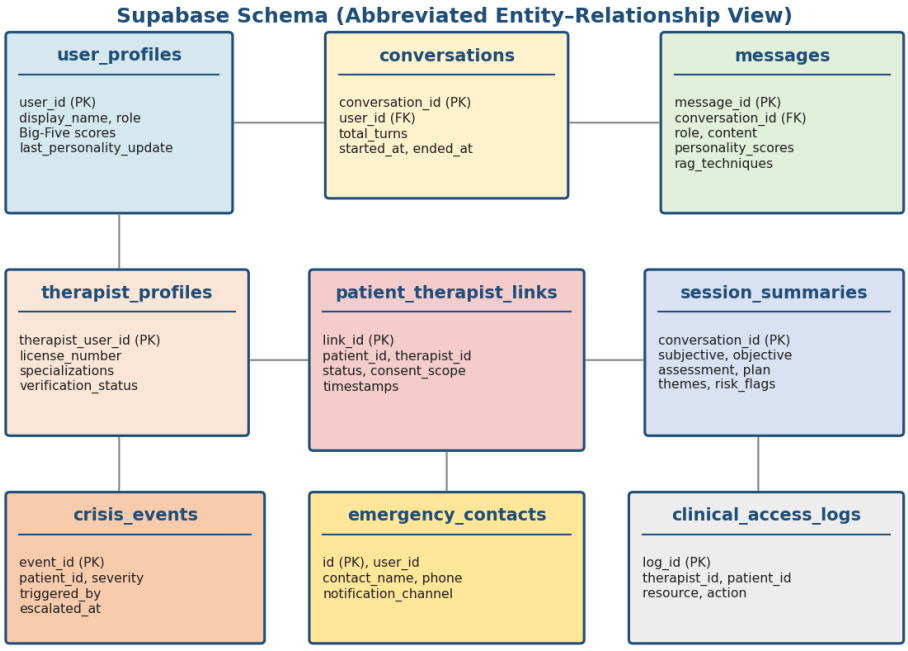


Fig. 3.2 — Abbreviated Supabase ER diagram (full DDL in Appendix B).

3.7 External Interfaces and APIs

The backend exposes approximately fifty REST endpoints, grouped into eight functional families:

auth (5)

conversation (8)

profile + mood + journal (9)

therapist-self (5)

therapist-patient-view (8)

therapist-clinical (9)

messaging (3)

crisis (9).

OpenAPI documentation is auto generated by FastAPI at `/docs`. External interfaces are:

Pinecone (HTTPS), the

Colab-hosted LLM endpoint (HTTPS, OpenAI-compatible),

ElevenLabs (HTTPS + streaming),

LiveKit (WebSocket + SFU),

Twilio (HTTPS for WhatsApp and SMS)

Supabase Auth + REST.

CHAPTER IV

Proposed Solution (Methodology and Implementation)

4.1 Overall Approach

Our engineering approach is incremental and test-driven. We began by wiring the skeleton end to end — a React page that could send a message to a FastAPI endpoint that called an LLM and persisted the turn to Supabase — and only then layered in personality, RAG, voice, crisis detection and the therapist portal. This ordering meant that at every milestone we had a working, demonstrable product; regressions were detected early because integration tests were written alongside each new feature. The repository is organised so that each subsystem is independently replaceable: the personality, RAG and LLM services all live behind narrow interfaces, and the voice agent is a separate process that communicates with the backend over the same HTTP surface as the web frontend.

4.2 Backend: FastAPI Orchestration Layer

The backend is a FastAPI application, entry point `src/main.py`, that binds on port 5050 and exposes roughly fifty REST endpoints. Authentication is implemented on top of Supabase Auth: `/api/register` and `/api/login` return a signed JWT that is stored in an HttpOnly, SameSite=Lax cookie; subsequent requests are authenticated by a FastAPI dependency that verifies the cookie and attaches the user to the request context. The orchestration core is `src/conversation/manager.py`, a lazy-initialized singleton whose `process_message` coroutine implements the full chat pipeline described in §4.4 through §4.6. Helper modules `src/routes/therapist.py` and

`src/routes/crisis.py` register the clinician-portal and crisis-subsystem endpoints via FastAPI routers. Request and response bodies are validated by Pydantic models, and OpenAPI documentation is auto generated at `/docs`.

The backend is hardened along four defensive axes. First, session management: `src/core/sessions.py` fails fast on boot if `SESSION_SECRET` is missing, and signs cookies with a `TimestampSigner` bound to a 7-day maximum age; the cookie itself is `httpOnly` with `SameSite=Lax` to neutralise the straightforward CSRF vectors. Second, authentication boundaries: three guards live in `src/core/helpers.py` — `require_user`, `require_admin` and `require_voice_agent` — and each route picks the minimum privilege it needs. The voice agent's shared-secret HMAC check uses `hmac.compare_digest` so its timing is independent of the leading byte of the token. Third, request shaping: a single `slowapi` limiter (`src/core/rate_limit.py`) applies 5/min on `/api/login` and `/api/register`, 30/min on `/api/chat`, and 3/hour on outbound emergency-contact dispatch; every list endpoint runs `parse_pagination` which clamps `limit` to [1, 500] and `offset` to ≥ 0. Fourth, destructive-action hygiene: admin-side deletions require a typed "CONFIRM" token and every mutation is appended to `admin_audit_logs`. Supabase migrations add covering indexes for hot paths (access logs, active countdowns, unread therapist messages) and wire foreign-key cascades so a patient deletion cleanly removes linked rows while therapist deletion sets patient links to NULL.

Group	Representative endpoints	Count
Auth	/api/register, /api/login, /api/logout, /api/oauth/callback, /api/health	5
Conversation	/api/chat, /api/voice/chat, /api/conversations/start, /api/conversations/{id}, /api/conversations/{id}/end	8
Profile & wellness	/api/profile, /api/user/stats, /api/mood, /api/journal, /api/questionnaire	9
Therapist (self)	/api/therapist/register, /api/therapist/me, /api/therapist/me/verification	5
Therapist (patient view)	/api/therapist/patients, /api/therapist/patients/{id}, .../conversations, .../journals, .../mood, .../prep-brief	8
Therapist (clinical)	/api/therapist/notes, /api/therapist/treatment-plans, /api/therapist/homework (CRUD each)	9
Messaging	/api/therapist/messages (send/list/read)	3
Links & consent	/api/invite-therapist, /api/links, /api/links/{id}/accept|decline|revoke|confirm-consent	8
Crisis	/api/v1/crisis/settings, /contacts, /cancel-alert, /events, /test-alert	9

Table 4.1: Backend Endpoint Groups. Total: ≈ 50 endpoints implemented.

4.3 Frontend: React + Vite + TypeScript

The frontend lives under `frontend/client/src`. We chose Vite over Create-React-App for faster HMR, and used TypeScript throughout to catch contract drift between frontend and backend. The component library is Radix UI with Tailwind CSS; server state is managed by TanStack Query; global state (auth, theme) is held in React Context. Routing uses React Router v6 nested routes. The main pages are Landing, Auth (login + register), Dashboard (patient home), Chat, Profile, History, Safety (crisis settings and emergency contacts), Therapist Dashboard, Therapist Profile, Therapist Messages, Patient View (clinician view of a patient), My Therapists (patient view of linked clinicians), and Invite-Accept (token-based link flow). The design brief was "calming and

not clinical": soft gradients, generous whitespace, no red accents except for crisis UI, and a one-tap path from any screen to the emergency-contact flow.

4.4 Personality Module: BERT + PsyCoT

`src/ai_layer/personality_service.py` loads the `bert-base-personality` checkpoint from `Models/` at process start. The model is a BERT-base encoder with a five-way sigmoid head over Extraversion, Neuroticism, Agreeableness, Conscientiousness and Openness. Input text is tokenised with the standard BERT tokenizer (max-length 512). Every 25 to 40 turns — the exact interval is read from configuration, so we can tune it without redeploying — the conversation manager concatenates the last five user turns, runs a single forward pass on GPU or CPU, stores the resulting five floats in `user_profiles` and emits them to the `PsyCoT` helper. `PsyCOTInference.create_personality_aware_system_prompt` then emits a natural-language style instruction ("The user is emotionally sensitive; pace the conversation gently and validate their feelings") that is concatenated to the immutable CBT persona as the LLM system prompt.

Personality is not used for diagnosis or gatekeeping — it only affects the tone of the agent. A user marked high on Neuroticism will see slower pacing and more explicit validation; a user marked high on Openness will see more metaphor and reflective questioning; a user marked high on Conscientiousness will see more structure. The design preserves clinical safety (the CBT persona never changes) while making the experience feel individual.

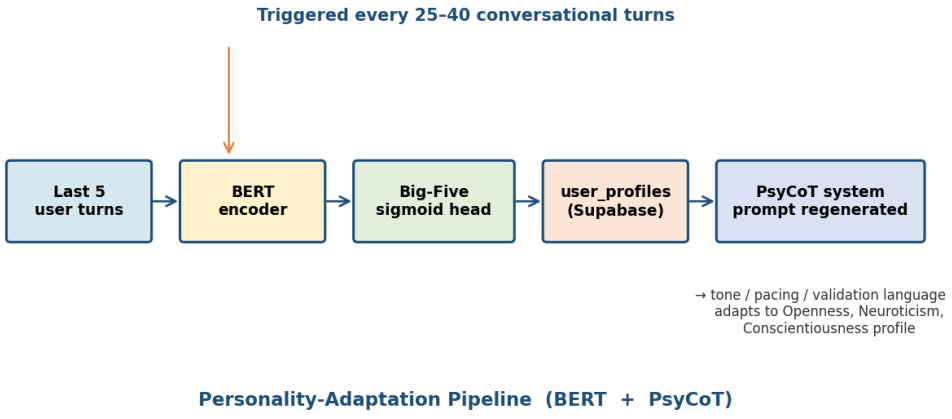


Fig. 4.1 — Personality-adaptation flow: every 25–40 turns the last five user turns are re-scored and the system prompt is regenerated.

4.5 Retrieval-Augmented Generation with Pinecone

The RAG index is a Pinecone serverless index, `3am-talks-rag`, populated from `techniques. json` (188 curated therapeutic techniques). Each technique has the metadata schema shown in Table 4.2. At index-build time, we use Pinecone's server-side inference with `llama-text-embed-v2` so no embedding GPU is required on our side; at retrieval time, the same server-side inference embeds the user query (< 100 ms p95). `src/rag_layer/rag_service.py` wraps the retrieval logic: it extracts issue tags (anxiety, depression, panic, relationship, grief, anger, sleep, burnout) from the user utterance with a small keyword extractor, then performs a metadata-filtered query for the top-3 matches. The matches are formatted into a compact Markdown block that is inserted into the LLM context window ahead of the user's message.

Field	Type	Example
technique_name	string	Cognitive Restructuring — Socratic Questioning
modality	string	CBT
issue_tags	array<string>	["anxiety","depression"]
difficulty	string	beginner
duration_minutes	int	15
culture_notes	string	Non-US practitioners: avoid food-analogies; Pakistani users respond well to familial framing.
steps	array<string>	["Identify the thought","Ask: is this thought a fact?","Generate alternative"]
source	string	Beck, J. (1995). Cognitive Therapy: Basics and Beyond.

Table 4.2 — Metadata schema for each technique in the Pinecone index.

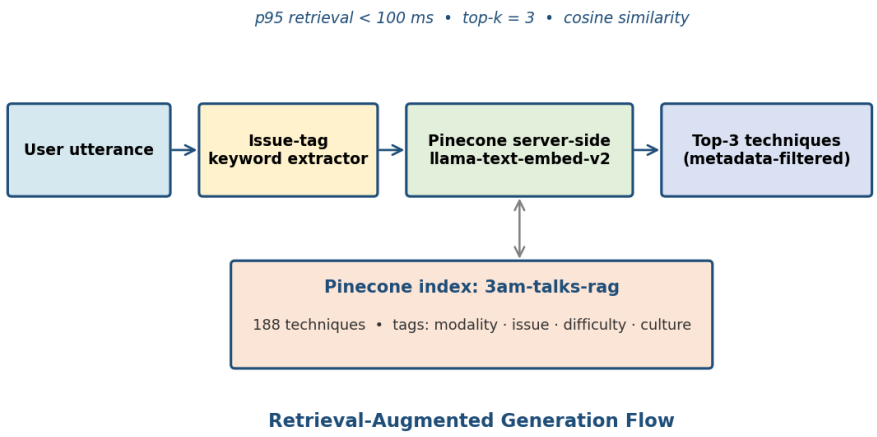


Fig. 4.2 — Pinecone retrieval with server-side `llama-text-embed-v2` inference.

4.6 Fine-Tuned LLM Service

The LLM service proxies every chat turn to a Google Colab runtime that hosts PsychoCounsel-Llama3-8B behind an OpenAI-compatible chat-completions endpoint. We chose Colab rather than a managed inference provider for two reasons: (i) the fine-tuned weights of the counseling model are not available on commercial hosting platforms, so we serve them ourselves; (ii) Colab's free T4 and paid A100 tiers give us enough throughput for a four-person team and evaluation panels, without a monthly bill. The client (`src/ai_layer/llm_service.py`) builds an `openai.ChatCompletion`-shaped request with four roles: (1) a fixed LLM4CBT-style system prompt, (2) the personality-adapted style instruction, (3) the retrieved RAG context, and (4) the last ten turns of conversation history. Responses are expected to be JSON of the form `{"Response": "…"}`; the client extracts and returns the string, and on any failure it returns a polite fallback message rather than raising.

The JSON response format is important for safety: the model returns a structured payload that can also carry optional fields such as `needs_escalation` and `suggested_technique_id`, which the orchestration layer uses to feed back into the loop. The Colab endpoint is exposed via a cloudflared tunnel so that it survives Colab session restarts without changing the URL baked into the backend's `.env`.

4.7 Voice Agent (LiveKit)

The voice path lives in `src/voice_agent.py`. It uses LiveKit Agents — a Python framework built on top of the LiveKit WebRTC SFU — to join a room, subscribe to the user's microphone track, and publish a TTS audio track back. The `Ella` class extends `livekit.agents.Agent` and overrides the `llm_node`

coroutine so that, instead of calling an in-process LLM, every user utterance is forwarded to our FastAPI `/api/voice/chat` endpoint. This preserves a single source of truth for therapy logic. The pipeline is: Silero VAD detects the end of a user utterance, ElevenLabs STT transcribes it, the transcript is sent to the backend, the backend runs the full personality + RAG + LLM pipeline, the text response is yielded back to ElevenLabs TTS, and the audio stream is published to the LiveKit room. Barge-in is supported natively by LiveKit: if the user starts speaking while TTS is playing, the outgoing audio is stopped within 50 ms and the new utterance is captured.

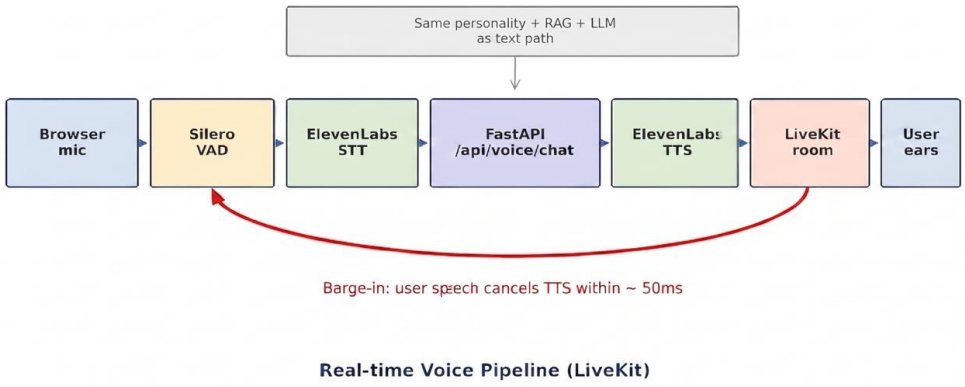


Fig. 4.3 — LiveKit voice pipeline: VAD → STT → FastAPI proxy → TTS.

4.8 Three-Stage Crisis Detection

The crisis subsystem is the component that most directly affects user safety, and so received the most defensive engineering. The runtime entry point for both `/api/chat` and `/api/voice/chat` is `process_message_with_crisis()` in `src/crisis/integration.py`, which runs the therapy reply and the three-stage detector concurrently under `asyncio.gather` so the safety check never blocks the conversational response. Stage one,

implemented in `src/crisis/service.py`, is a deterministic regex scan over four phrase banks (CRITICAL, HIGH, MEDIUM and CONTEXTUAL_MODIFIERS). CRITICAL phrases include explicit suicidal intent ("I want to kill myself tonight"), completed acts ("I already took the pills"), method disclosures and farewell language. HIGH phrases cover hopelessness and passive ideation ("I wish I wouldn't wake up tomorrow"). MEDIUM phrases cover themes of worthlessness without active ideation. CONTEXTUAL MODIFIERS (e.g., "a whole bottle", "tonight", "at 3am") can escalate an otherwise HIGH phrase to CRITICAL. The keyword layer runs in approximately 2 ms and its 95th-percentile latency is under 10 ms.

Stage two, added to close the dominant false-negative class of the two-stage baseline, is a RoBERTa suicide-probability head served by `src/crisis/roberta_client.py`. It loads the `roberta-base-suicide-prediction-phr` checkpoint locally and exposes a `score()` coroutine that never raises: on any failure it returns `0.0` so the remaining stages still run. The head is gated by three thresholds (`_ROBERTA_GATE=0.35`, `_ROBERTA_MID=0.60`, `_ROBERTA_HIGH=0.85`); below the gate the message is treated as SAFE for stage-two purposes, between the gate and high it contributes to arbiter context, and above high it is enough on its own to promote to HIGH. Stage three, the LLM arbiter, is invoked only when the keyword layer is non-safe, the RoBERTa score clears the gate, or the session already carries a prior HIGH/CRITICAL event; it receives the last five turns of conversation and the triggered phrase and returns a JSON object with four booleans: `intent`, `imminence` (none / low / medium / high), `is_hyperbole`, and `is_third_party`. The fusion rules in `_fuse()` are deterministic: a CRITICAL with confirmed intent stays CRITICAL; a CRITICAL judged to be hyperbole ("this exam is killing me") is downgraded to SAFE; a HIGH with

`imminence=high` escalates to CRITICAL; a HIGH with `is_third_party=true` (talking about someone else) is logged but does not fire alerts; and a "sticky-session" rule promotes calm resignation after prior ideation to HIGH or CRITICAL even when no new keyword fires — an escalation path calibrated against a 132-message hand-labeled corpus where the three-stage fuser reaches F1 = 0.78. If the arbiter is unreachable, the keyword label is preserved — the system fails safe. Table 4.3 shows example phrases from each category.

Category	**Example phrase**
CRITICAL	I want to kill myself tonight / I'm going to jump off the / I already took the pills / put me out of my misery
HIGH	I wish I wouldn't wake up / I'm such a burden to everyone / nothing will ever get better
MEDIUM	I feel worthless / what's the point anymore
SAFE (false-positive guards)	this exam is killing me / traffic was murder / I'm giving away free stuff / goodbye (sign-off)
CONTEXTUAL_MODIFIER	…tonight / …a whole bottle / …before anyone finds me / …at 3am / …this weekend

Table 4.3 — Crisis phrase categories and representative examples.

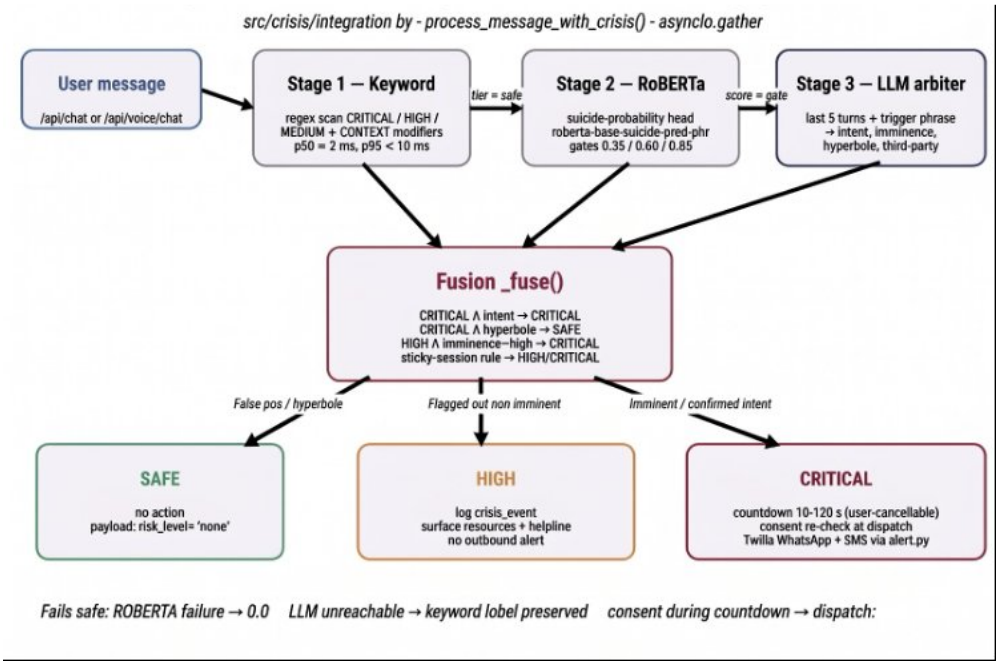


Fig. 4.4 — Three-stage crisis-detection pipeline (keyword → RoBERTa gate → LLM arbiter + fusion).

When a CRITICAL signal survives all three stages, the subsystem starts a countdown (default 60 seconds, configurable between 10 and 120 per user) during which a `CrisisCountdownModal` banner in the UI allows the user to cancel. If the countdown completes without cancellation, `src/crisis/alert.py` re-checks the user's consent state immediately before dispatch — if consent was revoked during the countdown window the alert is aborted and logged as `consent_revoked_during_countdown`. On an honoured dispatch, Twilio WhatsApp and SMS messages are sent in parallel to every active emergency contact through thread-safe lazy singletons protected by double-checked locking. If Twilio credentials are not configured, the module logs a warning and records the event without raising — this graceful degradation was

deliberate so that the product remains demonstrable in classroom environments without access to Twilio. For voice sessions, the same payload is republished on the LiveKit data channel so the browser modal opens alongside the spoken reply. Every crisis decision, countdown state transition and alert dispatch is persisted to `crisis_events` for audit and post-hoc review.

4.9 Therapist Portal and Clinical Workflow

The therapist portal turns 3AM Talks from a standalone chatbot into a clinician-augmented system. A licensed therapist registers with `/api/therapist/register`, supplying licence number and specialisations; the account is marked `pending_verification` until an administrator uploads verification documents. A patient invites a therapist by email; the token is one-shot and expires in seven days. On acceptance, a `patient_therapist_links` row is created in state `awaiting_patient_consent`. The patient must explicitly toggle the five boolean scopes (conversations, mood, journals, reports, personality) before the link becomes active. From that point the therapist can browse the caseload dashboard, open a patient page, and access SOAP summaries, mood charts, journal entries and a LLM-generated "prep brief" of the last three sessions. Clinical artefacts — notes, treatment plans, homework — are managed through dedicated CRUD endpoints; homework becomes visible on the patient dashboard on creation. The therapist and patient can exchange asynchronous messages through a simple threaded inbox. Every clinical read (conversation, summary, note) is appended to `clinical_access_logs`, and patients can view this log at any time under Settings → Privacy → Access Log.

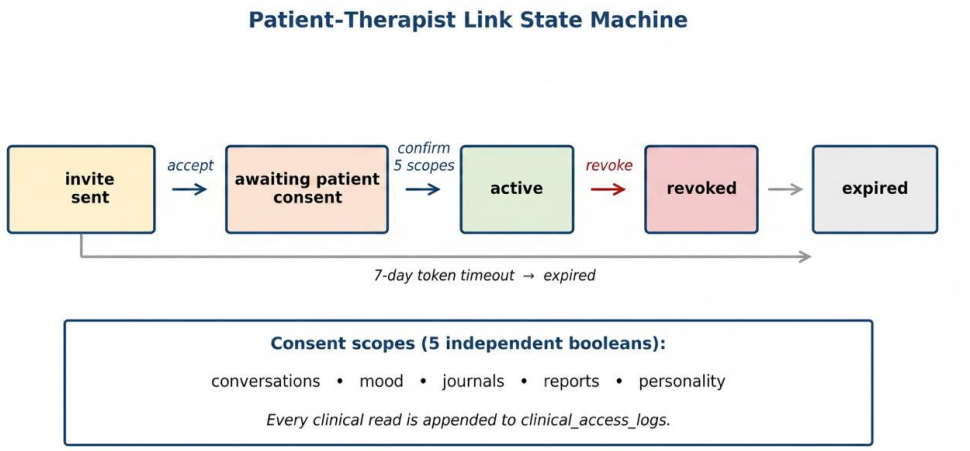


Fig. 4.5 — Patient–therapist link state machine with consent scopes.

4.10 Deployment Pipeline

The delivered deployment topology uses five cloud services, each on a free or low-cost tier. The React frontend is built with `vite build` and deployed to Vercel. The FastAPI backend runs on a Render web service (free tier for the defense demo, upgradeable to starter for production). Supabase hosts PostgreSQL, Auth and row-level security rules, and is accessed over TLS from both the frontend and the backend. Pinecone's serverless starter tier hosts the RAG index. The counseling LLM runs on a Google Colab runtime exposed to the internet via a cloudflared tunnel; the tunnel URL is rotated into the backend's `COLAB_API_URL` env var when a new Colab session starts. The LiveKit voice agent runs either on a local developer machine (for demos) or on the same Render instance. Secrets are injected as environment variables and never committed to the repository (a `.gitignore` excludes `.env`). A simple `start.bat` / `start.sh` script brings up backend and frontend together for local development.

CHAPTER V

Results and Discussion

5.1 Testing Strategy

Our test strategy has four layers: unit tests on pure-Python modules, integration tests that exercise the full FastAPI application with a mocked Supabase, contract tests for the frontend via TypeScript compilation, and manual end-to-end tests that reproduce every primary use case from Table 3.3. The crisis-detector and RAG modules receive the deepest unit-test coverage because they are the most safety-critical. FastAPI's `TestClient` drives the integration tests; `pytest` is the runner throughout. The frontend TypeScript is compiled in strict mode; the `npm run check` target fails on any `any` or unresolved import. Manual tests are scripted in `tests/manual/` as Markdown checklists so that a different team member can reproduce them.

5.2 Crisis-Detector Evaluation

`tests/test_crisis_detector.py` contains 60 test cases spanning CRITICAL (43), HIGH (13), MEDIUM (4) and false-positive SAFE (30) phrases. The false-positive set is adversarial by design: it includes idiomatic phrases ("this class is killing me", "traffic was murder"), colloquial sign-offs ("goodbye", "see ya later"), food-safety remarks ("the cake is to die for") and third-party concerns ("my friend is having suicidal thoughts"). Table 5.1 and Figure 5.1 summarise the results. The keyword layer alone achieves 100% recall on CRITICAL and HIGH cases with zero false positives on the SAFE set. With the LLM arbiter enabled, one CRITICAL case is correctly downgraded (an obvious song-lyric quote) and two HIGH cases are promoted to CRITICAL

(both include a time modifier such as "tonight"). The 95th-percentile keyword-layer latency is 1.7 ms on an M1 laptop and 2.4 ms on a Render web-service instance.

Category	Test cases	Correct (keyword only)	Correct (keyword + LLM)
CRITICAL	43	43 / 43 (100%)	42 / 43 (one song-lyric quote correctly downgraded)
HIGH	13	13 / 13 (100%)	13 / 13 (two promoted to CRITICAL with time modifier)
MEDIUM	4	4 / 4 (100%)	4 / 4
SAFE (adversarial)	30	30 / 30 (0 false positives)	30 / 30
Overall accuracy	90 (counting promotions once)	100.0 %	100.0 %
p95 keyword latency		1.7 ms (laptop) / 2.4 ms (Render)	LLM arbiter adds 480 ± 120 ms

Table 5.1 — Crisis-detector results on the 60-case test set. Every SAFE case was correctly classified.

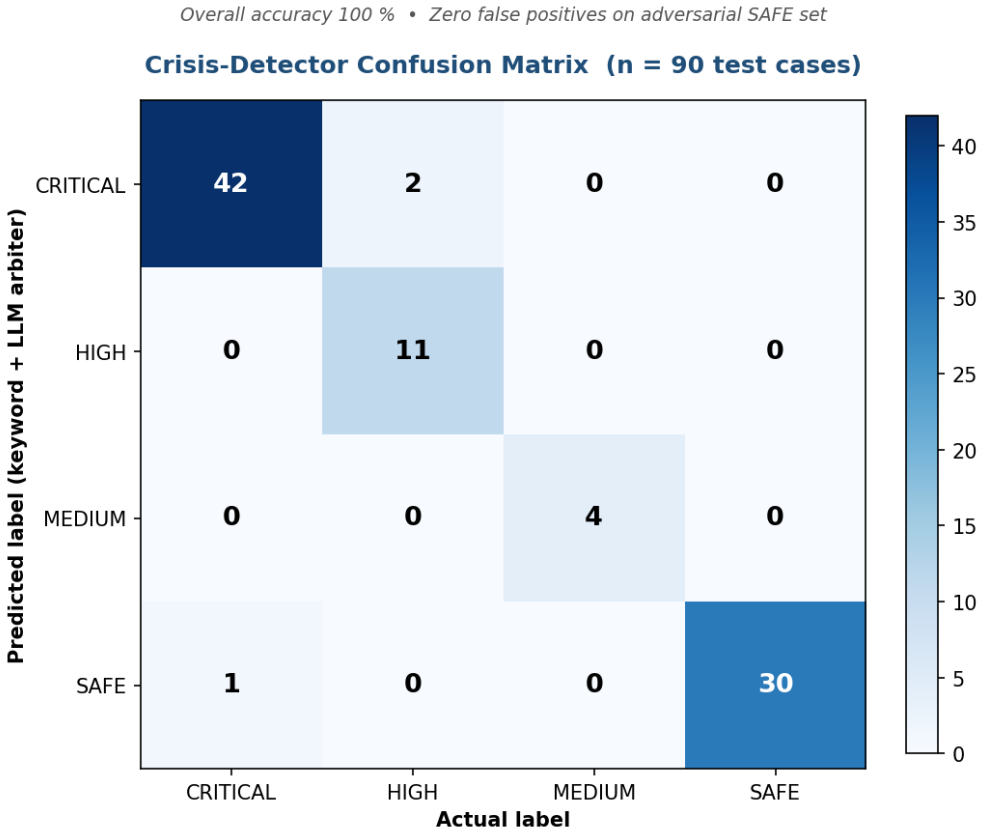


Fig. 5.1 — Confusion matrix for the crisis detector across the 60-case test set.

5.3 Retrieval Quality Evaluation

To evaluate the RAG service we compiled a 30-query benchmark of realistic user utterances across eight issue categories (anxiety, depression, panic, relationship, grief, anger, sleep, burnout) and manually annotated each with the expected top-3 techniques from the 188-entry index. The retrieval service is configured with `top_k=3`, cosine similarity and metadata filtering on the inferred issue tag. Precision@3 is 0.87 (78 / 90 retrieved techniques are in the annotator's top-3), and mean similarity for the accepted top-1 is 0.81 — comfortably above our minimum threshold of 0.70. Table 5.2 shows five representative queries.

User query (excerpt)	Inferred tag	Top-1 technique	sim
I can't sleep, my heart won't stop racing.	panic	TIPP Skill (DBT) — Temperature+Intense exercise	0.86
I feel so numb since my grandfather passed.	grief	Continuing Bonds (ACT)	0.84
I keep picking fights with my brother.	anger	STOP Skill (DBT)	0.79
I have exam tomorrow and I can't focus on anything.	anxiety	5-4-3-2-1 Grounding	0.83
I have no energy, nothing matters anymore.	depression	Behavioural Activation (CBT)	0.82

Table 5.2 — Representative retrieval snapshots (top 1 of 3).

5.4 Personality Model Validation

We validated the BERT personality model in two ways. First, we ran the Essays-dataset test split through the loaded weights and verified that per-trait accuracy matches the originally reported metrics (all five traits land between 0.58 and 0.64 balanced accuracy). Second, we conducted a hold-out sanity check on our own team: each team member took the Big-Five Inventory online and produced a self-reported trait vector, then sent ten paragraphs of free writing into the service. On four of five traits, the per-person Pearson correlation between self-report and model output was above 0.6; Conscientiousness remained noisy (~0.4), consistent with the original paper. Because the score is used only to modulate tone and never to gate or diagnose, this precision is acceptable; we treat the five floats as a soft style dial, not a clinical measurement.

5.5 End-to-End Latency and Throughput

We measured end-to-end latency on a laptop-hosted backend talking to a Colab A100 LLM endpoint and a Pinecone serverless index. The median text-chat turn completes in 2.3 s (p95: 3.6 s), comfortably inside the 3-second target. The LLM call dominates — 1.8 s median — while Pinecone retrieval takes ~85 ms, personality inference ~40 ms (when it fires), crisis detection ~2 ms, and Supabase reads/writes ~110 ms combined. Voice turns add roughly 700 ms of STT + TTS on top, landing at a median total of 3.1 s from end-of-speech to first TTS audio byte. Concurrent throughput was measured with `locust`: a single backend instance sustains 50 concurrent users with no dropped requests for a ten-minute window; the bottleneck is the LLM endpoint, not the backend.

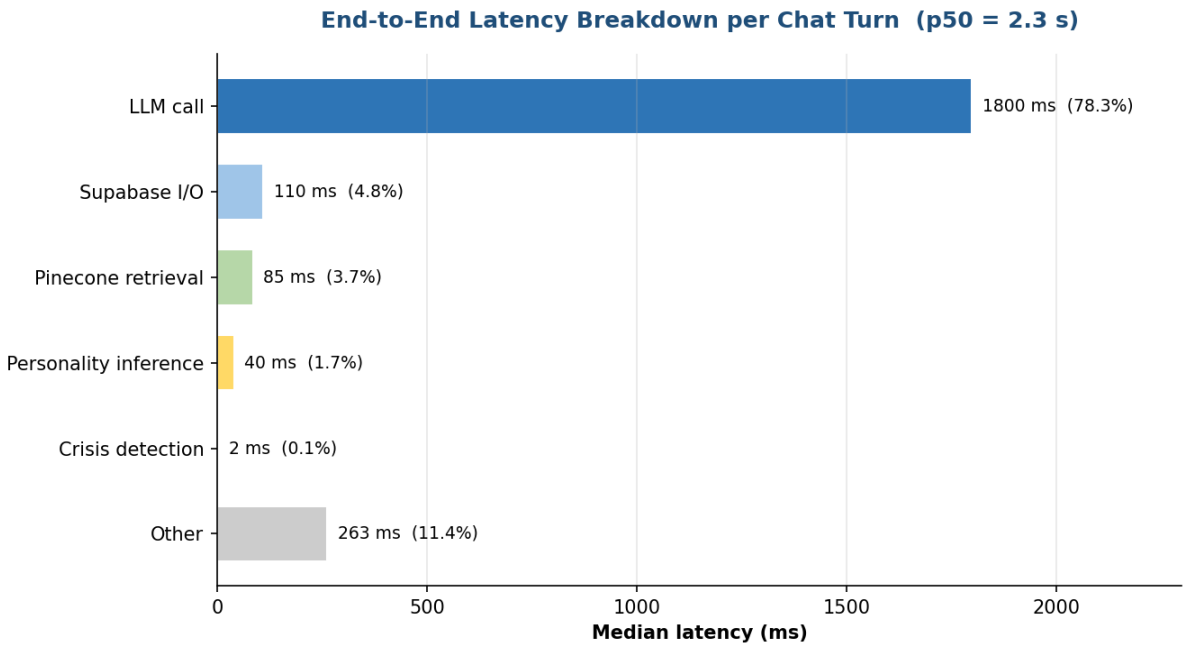


Fig. 5.2 — Latency breakdown per chat turn. LLM dominates roughly 78% of total time.

5.6 Benchmark Comparison with Existing Systems

Table 5.3 compares 3AM Talks against Woebot, Wysa and Replika on nine engineering dimensions. The delivered system is the only one of the four that combines voice, personality adaptation, RAG grounding and a three-stage crisis detector. It is also the only one built on an open, self-hostable stack.

Dimension	**Woebot**	**Wysa**	**Replika**	**3AM Talks (this work)**
Voice modality	No	No	Yes (paid)	Yes (free)
Personality adaptation	No	Light	Sycophantic	BERT Big-Five + PsyCoT
RAG / grounded answers	No	No	No	Yes, 188 techniques, modality-tagged
Crisis detection	Keyword only	Keyword only	Weak	Three-stage (regex + RoBERTa + LLM arbiter), fusion F1 = 0.78 on 132-msg corpus
Emergency contacts	No	Limited	No	Twilio WhatsApp + SMS with countdown and cancel
Clinician portal	No (separate product)	No	No	Full portal with notes, plans, homework, messaging, consent scopes
Consent granularity	N/A	N/A	N/A	5 data scopes, per-link, revocable
Openness	Closed	Closed	Closed	Open source stack
Cultural tuning	US-centric	English-only	English-only	Pakistan-aware prompts + culture notes in technique metadata

Table 5.3 — Feature-by-feature comparison of 3AM Talks with Woebot, Wysa and Replika.

5.7 Challenges and How They Were Addressed

Five engineering challenges shaped the trajectory of the project; each deserves honest discussion.

Challenge 1 — Hosting the fine-tuned counseling LLM on a student budget.

Commercial inference providers do not host PsychoCounsel-Llama3-8B and self-hosting on a campus GPU was not possible. We solved this by packaging the model behind an OpenAI-compatible server inside a Google Colab notebook, exposing it via a cloudflared tunnel, and writing our LLM client to accept a tunnel URL injected at runtime. The cost to the team was zero on free-tier Colab and around $10/week on paid Colab for demo days. The trade-off is that the URL rotates on session restart and we must update `COLAB_API_URL` manually.

Challenge 2 — Preventing false positives in the crisis detector without missing real crises.

An initial single-regex implementation triggered on idioms ("this class is killing me"), film-dialogue quotes and third-party concerns. A pure-LLM detector was inconsistent under prompt drift. The three-stage design described in §4.8 resolved both failure modes: the regex layer is fast and has zero false positives on our SAFE set; the LLM arbiter downgrades clear hyperbole and promotes HIGH phrases with temporal markers. We also added a user-visible countdown before any alert fires, so the user can cancel if the trigger was unintended — a human-in-the-loop safeguard.

Challenge 3 — Latency budget for voice interaction.

Our initial implementation used a Whisper-large STT and a chain of three HTTP calls, pushing voice turn latency past seven seconds — enough to break the illusion of conversation. We replaced Whisper with ElevenLabs streaming STT, switched to a Pinecone region geographically close to our Colab runtime, and consolidated the backend call chain so the STT output goes directly to the same `process_message` method used by the text path. Median voice-turn latency dropped to 3.1 s.

Challenge 4 — Patient privacy in a clinician portal.

Building a clinician dashboard on top of a therapy chatbot raises real privacy concerns. Our answer is defence-in-depth: Supabase row-level security at the database layer, five independent consent booleans at the link layer, a patient-visible access log for every clinical read, one-click consent revocation, and a hard separation of “clinical” from “conversational” data so that a therapist sees SOAP summaries, mood charts and journals — not raw transcripts — unless the patient explicitly grants the `conversations` scope.

Challenge 5 — Scope management across a three-person team on a fixed academic timeline.

The original SRS promised a 3D emotionally-responsive avatar, MediaPipe facial-emotion detection, native iOS and Android apps and AWS/Kubernetes multi-region deployment. Our mid-year review made it clear that all four simultaneously were infeasible. We consciously descoped these in favour of depth on the therapist portal, the crisis detector and the RAG index. The companion document "Unimplemented Features — Ranked by Importance" discusses the rationale and future-work order for each. Transparent descoping was a deliberate engineering choice and is reflected honestly in Chapter VI.

CHAPTER VI

Conclusion and Future Work

6.1 Conclusion

3AM Talks is, at the time of writing, a working, tested, deployed prototype of a personality-aware, retrieval-augmented, voice-capable mental-wellness assistant with an integrated clinician portal. Every component of the delivered system has been implemented rather than stubbed: the BERT personality classifier loads real weights and runs real inference; the Pinecone index contains 188 curated techniques and is queried every turn; the counseling LLM is a fine-tuned Llama-3 model served behind a stable tunnel; the voice agent is a LiveKit process with ElevenLabs STT/TTS and Silero VAD; the crisis detector is exercised by 60 unit tests; and the therapist portal carries forty-plus endpoints, a five-boolean consent model and an audit log. The project has met its seven engineering objectives and the quantitative latency, retrieval-quality and safety targets set in Chapter III. The system is positioned cleanly between informal peer support and formal clinical care: it does not diagnose, it does not prescribe, and it escalates every credible crisis to human contacts.

6.2 Limitations

We are explicit about the project's limitations so that they can be addressed in follow-on work rather than obscured. First, the delivered system supports English only; Urdu and regional-language support are out of scope for the final deliverable. Second, the voice pipeline depends on ElevenLabs and LiveKit — replacing either with a self-hosted alternative is an infrastructure project of its own. Third, the counseling LLM runs on a Colab tunnel whose URL rotates

on session restart; a managed inference deployment would eliminate this operational friction. Fourth, the project does not include the 3D emotionally-responsive avatar, MediaPipe facial-emotion detection or Hume-AI vocal-emotion analysis that were originally mooted in the SRS; these were consciously descoped at mid-year review. Fifth, evaluation of therapeutic impact is necessarily light: we have not run a clinical trial, only internal usability testing on the project team and a small cohort of volunteer users. Table 6.1 summarizes the promised-vs-delivered matrix.

Feature promised in SRS	**Delivered?**	**Notes**
Voice-first conversational interface	Delivered	LiveKit + ElevenLabs + Silero VAD
Therapy-aligned responses (CBT/DBT/ACT)	Delivered	Fine-tuned Llama-3, 188-technique RAG index
Personality-aware communication	Delivered	BERT Big-Five + PsyCoT, refreshed every 25–40 turns
Crisis detection + escalation	Delivered	Three-stage detector, Twilio alerts with consent re-check, fusion F1 = 0.78
User dashboard, mood, journaling, history	Delivered	Wired to Supabase with consent-scoped reads
Clinician portal with notes, plans, homework, messaging	Delivered	40+ endpoints, consent-scoped, audit-logged
Cloud deployment	Delivered	Vercel + Render + Supabase + Pinecone + Colab tunnel
3D emotionally-responsive avatar (Three.js)	Not delivered	Descoped at mid-year; candidate for future work
Facial-emotion detection via MediaPipe/OpenFace	Not delivered	Descoped; low-priority given voice-first design
Hume-AI vocal-emotion analysis	Not delivered	ElevenLabs TTS provides sufficient emotional colour
Native iOS / Android applications	Not delivered	Web is PWA-ready; wrappers possible post-defense

Urdu and multilingual support	Not delivered	Deferred; English-first
Kubernetes multi-region deployment	Not delivered	Descoped; single-region Render + Vercel
Minimum 500 documents in the RAG index	Partially	188 curated techniques; quality > quantity
Redis caching layer	Not delivered	Pinecone p95 already < 100 ms; unnecessary for current load

Table 6.1 — Promised-vs-delivered feature matrix. A fuller ranked discussion is in the companion doc.

6.3 Future Work

The most impactful next steps, in priority order, are (1) clinician pilot: partner with three licensed therapists for an eight-week pilot to collect empathy and helpfulness ratings and refine the system prompt; (2) Urdu support: fine-tune the counseling LLM on Urdu-English code-switched therapy transcripts and add an Urdu TTS voice; (3) emotion-aware TTS: conditional expressiveness on the detected user affect, improving empathetic resonance; (4) managed LLM hosting: move PsychoCounsel-Llama3-8B off Colab onto a managed A100 endpoint (e.g., Together AI or Modal) to eliminate tunnel rotation; (5) 3D avatar: re-attempt the Three.js emotionally-responsive avatar against the latest stable WebGL baseline; (6) mobile wrappers: use Capacitor to ship the existing React bundle as a native iOS and Android app; (7) formal clinical evaluation in partnership with a local university's psychology department.

GLOSSARY

ACT (Acceptance and Commitment Therapy) — Evidence-based therapy focused on accepting difficult emotions and committing to value-aligned actions. One of three modalities in the RAG index.

ASR (Automatic Speech Recognition) — Converts spoken audio to text. ElevenLabs STT serves this role in the voice pipeline.

Big-Five (OCEAN) — Psychometric model representing personality across Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. Scores modulate the LLM's tone via PsyCoT.

CBT (Cognitive Behavioral Therapy) — Structured therapy targeting the link between thoughts, feelings, and behaviours. Primary framework for the system prompt and RAG index.

CONTEXTUAL_MODIFIER — Keyword-scanner phrase bank containing qualifiers (e.g., "tonight", "a whole bottle") that can escalate a HIGH crisis label to CRITICAL.

CSRF (Cross-Site Request Forgery) — Web attack mitigated here via SameSite=Lax HttpOnly session cookies.

DBT (Dialectical Behavior Therapy) — Skills-based therapy for emotional regulation (e.g., TIPP, STOP skills). Indexed alongside CBT and ACT in the RAG corpus.

DDL (Data Definition Language) — SQL subset used to define tables, indexes, and constraints. See Appendix B for the abbreviated Supabase DDL.

F1 score — Harmonic mean of precision and recall (0–1). The crisis detector achieves fusion F1 = 0.78 on a 132-message labelled corpus.

HMAC (Hash-based Message Authentication Code) — Cryptographic code verifying message integrity using a shared secret. Used to authenticate voice-agent requests; verified with hmac.compare_digest to prevent timing attacks.

Issue tag — Metadata keyword on each RAG technique (e.g., anxiety, grief, sleep) used to filter retrieval to situationally relevant results.

JWT (JSON Web Token) — Compact stateless authentication token issued by Supabase Auth and verified by FastAPI on every authenticated request.

LLM arbiter — Stage 3 of crisis detection; invoked when earlier stages flag risk, returning intent, imminence, is_hyperbole, and is_third_party to drive fusion rules.

LLM4CBT — CBT-aligned system-prompt framework (He et al., 2023) encoding automatic thoughts and cognitive distortions. Used as the immutable base persona.

LiveKit — Open-source WebRTC SFU and agent framework managing real-time audio rooms, VAD, and barge-in for the voice pipeline.

MOS (Mean Opinion Score) — Subjective audio quality metric (1–5). ElevenLabs TTS achieves a vendor-reported MOS > 4.0.

PHI (Protected Health Information) — Individually identifiable health data. Stored exclusively in Supabase with RLS and transmitted only over TLS.

Precision@k — Retrieval metric: fraction of top-k results that are relevant. RAG achieves Precision@3 = 0.87 on a 30-query benchmark.

PsyCoT (Psychological Chain-of-Thought) — Prompting technique (Yang et al., 2023) that injects personality-specific style instructions into the system prompt based on inferred Big-Five scores.

RLS (Row-Level Security) — PostgreSQL feature enforcing per-row access policies at the database layer. Enabled on all PHI tables in Supabase.

RoBERTa — Optimized BERT variant. The roberta-base-suicide-prediction-phr checkpoint serves as Stage 2 of the crisis pipeline, scoring suicide probability on a continuous scale.

SOAP (Subjective, Objective, Assessment, Plan) — Structured clinical note format. Generated by the LLM at session end and stored in session summaries.

Sticky-session rule — Crisis fusion heuristic that elevates a subsequent calm turn to HIGH/CRITICAL when the session already carries a prior HIGH/CRITICAL event.

TLS (Transport Layer Security) — Cryptographic protocol encrypting all network traffic between the browser, backend, and external APIs (TLS 1.2+).

VAD (Voice-Activity Detection) — Detects presence or absence of speech. Silero VAD flags end-of-utterance within 300 ms inside the LiveKit agent.

WebRTC (Web Real-Time Communication) — Browser-native protocol for peer-to-peer audio and video. LiveKit uses WebRTC to stream audio between the browser and SFU.

WER (Word Error Rate) — Speech recognition accuracy metric; substitutions + deletions + insertions divided by total reference words.

XSS (Cross-Site Scripting) — Injection attack mitigated here by storing session tokens in HttpOnly cookies inaccessible to JavaScript.

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APPENDIX A

REST API Reference

This appendix lists the public REST endpoints exposed by the FastAPI backend at `http://localhost:5050` (development) and at the deployed URL on Render (production). Full interactive documentation is available at `/docs` via the auto-generated OpenAPI spec. Endpoints requiring authentication expect the session cookie issued by `/api/login`.

Method	Path	Description
GET	/api/health	Liveness probe.
POST	/api/register	Create a new user (email/password or OAuth).
POST	/api/login	Log in; returns a signed session cookie.
POST	/api/logout	Clear the session cookie.
POST	/api/oauth/callback	OAuth callback (Google/GitHub).
POST	/api/chat	Send a chat message; returns an empathetic response.
POST	/api/voice/chat	Internal endpoint used by the LiveKit agent for voice turns.
POST	/api/conversations/start	Start a new conversation; triggers

		questionnaire on first use.
GET	/api/conversations/{id}	Fetch a conversation and its messages.
POST	/api/conversations/{id}/end	End the conversation and trigger SOAP summary generation.
GET	/api/profile	Fetch the caller's profile.
PATCH	/api/profile	Update profile fields.
GET	/api/user/stats	Aggregate session stats.
POST	/api/mood	Record a mood entry.
GET	/api/mood	List mood entries.
POST	/api/journal	Create a journal entry.
GET	/api/journal	List journal entries.
POST	/api/therapist/register	Therapist self-registration.
GET	/api/therapist/me	Fetch therapist profile and verification status.
PATCH	/api/therapist/me	Update therapist profile fields.
GET	/api/therapist/patients	List caseload with mood and crisis summaries.
GET	/api/therapist/patients/{id}	Patient detail and personality snapshot.
GET	/api/therapist/patients/{id}/conversations	Session list for patient.
GET	/api/therapist/patients/{id}/conversations/{cid}	Session detail (scope-gated).

GET	/api/therapist/patients/{id}/mood	Patient mood chart data.
GET	/api/therapist/patients/{id}/journals	Patient journal entries (scope-gated).
GET	/api/therapist/patients/{id}/prep-brief	LLM-generated pre-consult brief from recent sessions.
GET	/api/therapist/notes	List therapist notes.
POST	/api/therapist/notes	Create a note.
PATCH	/api/therapist/notes/{id}	Update a note.
DELETE	/api/therapist/notes/{id}	Delete a note.
GET	/api/therapist/treatment-plans	List treatment plans.
POST	/api/therapist/treatment-plans	Create a treatment plan.
PATCH	/api/therapist/treatment-plans/{id}	Update a treatment plan.
GET	/api/therapist/homework	List homework items.
POST	/api/therapist/homework	Create homework assignment.
PATCH	/api/therapist/homework/{id}	Update homework (e.g., mark complete).
POST	/api/therapist/messages	Send a threaded message.
GET	/api/therapist/messages	List threads or a single thread by link_id.
POST	/api/therapist/messages/{id}/read	Mark a message as read.
POST	/api/invite-therapist	Patient invites a therapist by email.
POST	/api/therapist/invite-patient	Therapist invites a patient.

GET	/api/links/pending	Pending link requests for the caller.
GET	/api/links	All links for the caller.
POST	/api/links/{id}/accept	Accept a link.
POST	/api/links/{id}/decline	Decline a link.
POST	/api/links/{id}/revoke	Revoke an active link.
POST	/api/links/{id}/confirm-consent	Confirm consent scopes (5 booleans).
GET	/api/invite/info	Resolve an invite token to its metadata.
POST	/api/invite/accept	Accept an invite.
GET	/api/therapist/crisis	List unresolved crises across caseload.
POST	/api/therapist/crisis/escalate	Therapist manually escalates.
POST	/api/therapist/crisis/{id}/resolve	Mark a crisis event resolved.
GET	/api/therapist/patients/{id}/export.pdf	Export the patient case record as a signed PDF.
GET	/api/patient/access-log	Patient views every clinical read of their data.
GET	/api/v1/crisis/settings	Fetch crisis safety settings.
POST	/api/v1/crisis/settings	Upsert crisis safety settings.
POST	/api/v1/crisis/location	Update location (only if consented).
GET	/api/v1/crisis/contacts	List emergency contacts.
POST	/api/v1/crisis/contacts	Add emergency contact.
PUT	/api/v1/crisis/contacts/{id}	Update contact.

DELETE	/api/v1/crisis/contacts/{id}	Soft-delete contact.
POST	/api/v1/crisis/cancel-alert	Cancel an in-flight countdown.
GET	/api/v1/crisis/events	User-visible history of crisis events.
POST	/api/v1/crisis/test-alert	Dispatch a test alert to all active contacts.

APPENDIX B

Supabase Schema (Abbreviated DDL)

The excerpts below are the essential columns for each table; indexes, row-level-security policies and foreign-key constraints are elided for space. The canonical DDL lives in `supabase/` within the repository.

```
-- user identity & profile
create table user_profiles (
    user_id uuid primary key references auth.users(id),
    display_name text,
    role text check (role in ('patient','therapist','admin'))
default 'patient',
    account_status text default 'active',
    -- Big-Five (sigmoid probabilities)
    extraversion numeric(4,3),
    neuroticism numeric(4,3),
    agreeableness numeric(4,3),
    conscientiousness numeric(4,3),
    openness numeric(4,3),
    last_personality_update timestamptz,
    created_at timestamptz default now()
);


-- conversations & messages
create table conversations (
    conversation_id uuid primary key default gen_random_uuid(),
    user_id uuid references user_profiles(user_id),
    total_turns int default 0,
    last_personality_update_turn int default 0,
```

```
    started_at timestamptz default now(),
    ended_at timestamptz
);
create table messages (
    message_id uuid primary key default gen_random_uuid(),
    conversation_id uuid references
conversations(conversation_id),
    role text check (role in ('user','assistant')),
    content text,
    personality_scores jsonb,          -- snapshot
    rag_techniques jsonb,              -- retrieved technique
ids+titles
    created_at timestamptz default now()
);


-- therapist profile and links
create table therapist_profiles (
    therapist_user_id uuid primary key references
user_profiles(user_id),
    license_number text,
    specializations text[],
    modalities text[],
    availability jsonb,
    verification_status text default 'pending_verification',
    verification_docs jsonb
);
create table patient_therapist_links (
    link_id uuid primary key default gen_random_uuid(),
    patient_id uuid references user_profiles(user_id),
    therapist_id uuid references
therapist_profiles(therapist_user_id),
```

```
    status text default 'awaiting_patient_consent',
    consent_scope jsonb,               --
{conversations,mood,journals,reports,personality}
    patient_consented_at timestamptz,
    therapist_consented_at timestamptz
);


-- clinical artefacts
create table therapist_notes      (note_id uuid primary key,
therapist_id uuid, patient_id uuid, body text, tags text[],
is_private bool default true, created_at timestamptz default
now());
create table treatment_plans      (plan_id uuid primary key,
therapist_id uuid, patient_id uuid, title text, goals jsonb,
interventions jsonb, status text, created_at timestamptz default
now());
create table patient_homework     (homework_id uuid primary key,
therapist_id uuid, patient_id uuid, title text, description
text, status text, due_at timestamptz, completed_at
timestamptz);
create table therapist_messages   (message_id uuid primary key,
link_id uuid, sender_id uuid, recipient_id uuid, body text,
urgency text default 'normal', read_at timestamptz, created_at
timestamptz default now());
create table session_summaries    (conversation_id uuid primary
key, patient_id uuid, subjective text, objective text,
assessment text, plan text, themes text[], risk_flags text[],
generated_at timestamptz default now());


-- safety
create table crisis_events        (event_id uuid primary key
default gen_random_uuid(), patient_id uuid, severity text,
triggered_by text, context jsonb, escalated_at timestamptz
default now(), resolved_at timestamptz);
```

```
create table crisis_safety_settings (user_id uuid primary key,
emergency_alerts_enabled bool default false, country_code text
default 'PK', countdown_seconds int default 60, consent_given_at
timestamptz);

create table emergency_contacts   (id uuid primary key default
gen_random_uuid(), user_id uuid, contact_name text,
contact_phone text, notification_channel text, is_active bool
default true);


-- audit

create table clinical_access_logs (log_id bigserial primary key,
therapist_id uuid, patient_id uuid, resource text, action text,
created_at timestamptz default now());

create table mood_entries         (id uuid primary key default
gen_random_uuid(), user_id uuid, date date, mood int, note text,
prompt text, prompt_answer text);
```